Portafolio PI

2025-08-04

Librerías

```
library(rvest)
library(dplyr)
library(BatchGetSymbols)
library(ggplot2)
library(tidyr)
library(rlang)
library(quadprog)
library(readxl)
library(ggrepel)
library(knitr)
library(RColorBrewer)
library(ggthemes)
library(xtable)
library(lubridate)
library(tidyquant)
library(foreach)
library(PortfolioAnalytics)
library(quantmod)
library(PerformanceAnalytics)
library(timetk)
library(ggcorrplot)
library(ggrepel)
library(cluster)
library(factoextra)
library(caret)
```

Descarga de acciones

Se descarga la información de las acciones de interes.

Asímismo, se selecciona la fecha de inicio, última fecha, frecuencia, el nombre de cada ticker y se guarda en un dataframe:

```
primera_fecha <- "2025-07-01"
ultima_fecha <- "2025-12-31"

frecuencia <- "daily"

tickers <- c("AC.MX", "ALFAA.MX", "ALSEA.MX", "AMXB.MX", "ASURB.MX", "BBAJIOO.MX",
             "BIMBOA.MX", "BOLSAA.MX","CEMEXCPO.MX", "CHDRAUIB.MX", "CUERVO.MX",
```

```r
             "FEMSAUBD.MX","GAPB.MX", "GCARSOA1.MX", "GCC.MX", "GENTERA.MX",
             "GFINBURO.MX", "GFNORTEO.MX", "GMEXICOB.MX", "GRUMAB.MX", "KIMBERA.MX",
             "LABB.MX","LIVEPOLC-1.MX", "MEGACPO.MX", "OMAB.MX", "ORBIA.MX",
             "PINFRA.MX", "Q.MX", "RA.MX", "TLEVISACPO.MX", "VESTA.MX", "WALMEX.MX")

descarga <- BatchGetSymbols(tickers = tickers,
                            first.date = primera_fecha,
                            last.date = ultima_fecha,
                            freq.data = frecuencia)
```

```
##
## Running BatchGetSymbols for:
##    tickers =AC.MX, ALFAA.MX, ALSEA.MX, AMXB.MX, ASURB.MX, BBAJIOO.MX, BIMBOA.MX, BOLSAA.MX, CEMEXCPO
##    Downloading data for benchmark ticker
## ^GSPC | yahoo (1|1) | Not Cached | Saving cache
## AC.MX | yahoo (1|32) | Not Cached | Saving cache - Got 98% of valid prices | You got it!
## ALFAA.MX | yahoo (2|32) | Not Cached - Error in download..
## ALSEA.MX | yahoo (3|32) | Not Cached | Saving cache - Got 98% of valid prices | Boa!
## AMXB.MX | yahoo (4|32) | Not Cached | Saving cache - Got 98% of valid prices | Well done!
## ASURB.MX | yahoo (5|32) | Not Cached | Saving cache - Got 98% of valid prices | Looking good!
## BBAJIOO.MX | yahoo (6|32) | Not Cached | Saving cache - Got 98% of valid prices | Feels good!
## BIMBOA.MX | yahoo (7|32) | Not Cached | Saving cache - Got 98% of valid prices | Looking good!
## BOLSAA.MX | yahoo (8|32) | Not Cached | Saving cache - Got 98% of valid prices | Good stuff!
## CEMEXCPO.MX | yahoo (9|32) | Not Cached | Saving cache - Got 98% of valid prices | Looking good!
## CHDRAUIB.MX | yahoo (10|32) | Not Cached | Saving cache - Got 98% of valid prices | Got it!
## CUERVO.MX | yahoo (11|32) | Not Cached | Saving cache - Got 98% of valid prices | Feels good!
## FEMSAUBD.MX | yahoo (12|32) | Not Cached | Saving cache - Got 98% of valid prices | You got it!
## GAPB.MX | yahoo (13|32) | Not Cached | Saving cache - Got 98% of valid prices | Well done!
## GCARSOA1.MX | yahoo (14|32) | Not Cached | Saving cache - Got 98% of valid prices | Well done!
## GCC.MX | yahoo (15|32) | Not Cached | Saving cache - Got 98% of valid prices | OK!
## GENTERA.MX | yahoo (16|32) | Not Cached | Saving cache - Got 98% of valid prices | Well done!
## GFINBURO.MX | yahoo (17|32) | Not Cached | Saving cache - Got 98% of valid prices | You got it!
## GFNORTEO.MX | yahoo (18|32) | Not Cached | Saving cache - Got 98% of valid prices | Feliz que nem lar
## GMEXICOB.MX | yahoo (19|32) | Not Cached | Saving cache - Got 98% of valid prices | Feels good!
## GRUMAB.MX | yahoo (20|32) | Not Cached | Saving cache - Got 98% of valid prices | Looking good!
## KIMBERA.MX | yahoo (21|32) | Not Cached | Saving cache - Got 98% of valid prices | Mas bah tche, que
## LABB.MX | yahoo (22|32) | Not Cached | Saving cache - Got 98% of valid prices | Good job!
## LIVEPOLC-1.MX | yahoo (23|32) | Not Cached | Saving cache - Got 98% of valid prices | Nice!
## MEGACPO.MX | yahoo (24|32) | Not Cached | Saving cache - Got 98% of valid prices | Nice!
## OMAB.MX | yahoo (25|32) | Not Cached | Saving cache - Got 98% of valid prices | Nice!
## ORBIA.MX | yahoo (26|32) | Not Cached | Saving cache - Got 98% of valid prices | Good job!
## PINFRA.MX | yahoo (27|32) | Not Cached | Saving cache - Got 98% of valid prices | You got it!
## Q.MX | yahoo (28|32) | Not Cached | Saving cache - Got 98% of valid prices | Nice!
## RA.MX | yahoo (29|32) | Not Cached | Saving cache - Got 98% of valid prices | OK!
## TLEVISACPO.MX | yahoo (30|32) | Not Cached | Saving cache - Got 98% of valid prices | Feels good!
## VESTA.MX | yahoo (31|32) | Not Cached | Saving cache - Got 98% of valid prices | Good job!
## WALMEX.MX | yahoo (32|32) | Not Cached | Saving cache - Got 98% of valid prices | Mais contente que
```

```r
precios <- descarga$df.tickers
```

Verificación de acciones descargadas

```r
empresas <- precios %>% dplyr::select(ticker) %>% unique()
empresas <- empresas$ticker
empresas
```

```
##  [1] "AC.MX"         "ALSEA.MX"      "AMXB.MX"       "ASURB.MX"
##  [5] "BBAJIOO.MX"    "BIMBOA.MX"     "BOLSAA.MX"     "CEMEXCPO.MX"
##  [9] "CHDRAUIB.MX"   "CUERVO.MX"     "FEMSAUBD.MX"   "GAPB.MX"
## [13] "GCARSOA1.MX"   "GCC.MX"        "GENTERA.MX"    "GFINBURO.MX"
## [17] "GFNORTEO.MX"   "GMEXICOB.MX"   "GRUMAB.MX"     "KIMBERA.MX"
## [21] "LABB.MX"       "LIVEPOLC-1.MX" "MEGACPO.MX"    "OMAB.MX"
## [25] "ORBIA.MX"      "PINFRA.MX"     "Q.MX"          "RA.MX"
## [29] "TLEVISACPO.MX" "VESTA.MX"      "WALMEX.MX"
```

Se acomoda la información en un dataframe

```r
df_precios <- precios %>%
  dplyr::select(ref.date, price.adjusted, ticker) %>%
  pivot_wider(names_from = ticker, values_from = price.adjusted)

names(df_precios)[1] <- "Fecha"

num_acciones <- length(names(df_precios)) -1
nombres_acciones <- names(df_precios)[2:(num_acciones+1)]
nombres_genericos <- paste0("accion_", 1:num_acciones)
nombres_originales <- names(df_precios)[2:(num_acciones+1)]
names(df_precios)[2:(num_acciones+1)] <- nombres_genericos

for(i in 1:num_acciones){
  columna <- as.name(paste0("accion_",i))
  vector_precios <- dplyr::pull(df_precios[columna], columna)
  log_rendimientos <- c()
  for(j in 2:length(vector_precios)){
    log_rendimientos[j] <- log(vector_precios[j]/vector_precios[j-1])
  }
  df_precios <- df_precios %>% mutate(!! as.name(paste0("rendimiento_",i)) := log_rendimientos)
}
```

Se guardan los log-rendimientos de cada acción por fecha y sin fecha

```r
df_rendims_acciones <- df_precios %>% dplyr::select(Fecha, starts_with("rendimiento_")) %>%
  drop_na()

df_rendimientos <- df_precios %>% dplyr::select(starts_with("rendimiento_")) %>% drop_na()
```

Se guarda la correspondencia del nombre de las acciones con el nombre genérico

```r
df_nombres <- data.frame(nombres_originales = nombres_originales,
                         nombres_genericos = nombres_genericos,
                         rendims_genericos = names(df_rendimientos))
```

Construcción de portafolio

De la pagina de Banxico, se obtiene el historico de las tasas cetes

```r
historico_cete <- read.csv("/Users/Arturo/Desktop/Cetes_enero18-nov24.csv")
```

Se ajusta el rendimiento libre de riesgo para datos semanales

```r
r_f <- ((mean(historico_cete$Valores) / 100 + 1)^(1 / 360)) - 1  # Convertir r_f diario
```

##Construcción de portafolio óptimo usando PCA y clustering

MATRIZ DE RENDIMIENTOS Y ESTADÍSTICAS BASE

```r
# Convertir rendimientos a matriz
mat_rendimientos <- as.matrix(df_rendimientos)

# Estadísticas por acción
medias      <- colMeans(mat_rendimientos)
covarianza  <- cov(mat_rendimientos)
desv_est    <- apply(mat_rendimientos, 2, sd)

# Sharpe individual (anualizado 252 días)
sharpe_ind <- (medias * 252 - r_f * 252) / (desv_est * sqrt(252))

df_stats <- data.frame(
  accion        = nombres_originales,
  media_diaria  = medias,
  desv_diaria   = desv_est,
  sharpe        = sharpe_ind
)
print(df_stats)
```

```
##                      accion  media_diaria desv_diaria       sharpe
## rendimiento_1         AC.MX -8.464945e-05  0.01359931 -0.101340471
## rendimiento_2       ALSEA.MX  5.352847e-04  0.02017190  0.419543423
## rendimiento_3        AMXB.MX  1.120105e-03  0.01360828  1.304112082
## rendimiento_4       ASURB.MX  1.669384e-04  0.01335197  0.195901598
## rendimiento_5     BBAJIOO.MX  4.504672e-04  0.01749544  0.406766208
## rendimiento_6      BIMBOA.MX  9.073609e-04  0.01743013  0.824406387
## rendimiento_7      BOLSAA.MX -1.199153e-03  0.01600157 -1.191780397
## rendimiento_8    CEMEXCPO.MX  3.629959e-03  0.01810955  3.180058348
## rendimiento_9    CHDRAUIB.MX -1.426970e-03  0.01410847 -1.608028836
## rendimiento_10     CUERVO.MX -8.111496e-04  0.02321902 -0.556052446
## rendimiento_11   FEMSAUBD.MX -2.366855e-04  0.01578239 -0.240246244
## rendimiento_12       GAPB.MX  8.130231e-04  0.01684680  0.764059163
## rendimiento_13   GCARSOA1.MX -1.138854e-03  0.02240061 -0.808600167
```

```
## rendimiento_14        GCC.MX -2.926904e-05  0.01605764 -0.031077050
## rendimiento_15    GENTERA.MX  9.310000e-04  0.02353511  0.626501266
## rendimiento_16   GFINBURO.MX -1.044206e-03  0.01601099 -1.037453063
## rendimiento_17   GFNORTEO.MX  4.523280e-05  0.01501082  0.045544204
## rendimiento_18   GMEXICOB.MX  3.320510e-03  0.02389265  2.204739365
## rendimiento_19     GRUMAB.MX -2.930465e-04  0.01123673 -0.417057520
## rendimiento_20    KIMBERA.MX  1.093564e-03  0.01616513  1.071775540
## rendimiento_21       LABB.MX -9.180868e-04  0.02034604 -0.718005415
## rendimiento_22 LIVEPOLC-1.MX  6.099345e-05  0.01161646  0.080390118
## rendimiento_23    MEGACPO.MX -1.870342e-04  0.01999219 -0.150232032
## rendimiento_24       OMAB.MX -4.994180e-05  0.01715120 -0.048229527
## rendimiento_25      ORBIA.MX  1.446997e-03  0.02844398  0.806355654
## rendimiento_26     PINFRA.MX  1.861316e-03  0.01954525  1.509987304
## rendimiento_27          Q.MX  1.097861e-05  0.02200729  0.006356416
## rendimiento_28         RA.MX -7.261054e-04  0.02036535 -0.567677933
## rendimiento_29 TLEVISACPO.MX  1.612922e-03  0.03024001  0.845566666
## rendimiento_30      VESTA.MX  5.344038e-04  0.01353433  0.624264830
## rendimiento_31     WALMEX.MX -6.322323e-04  0.01609427 -0.625736287
```

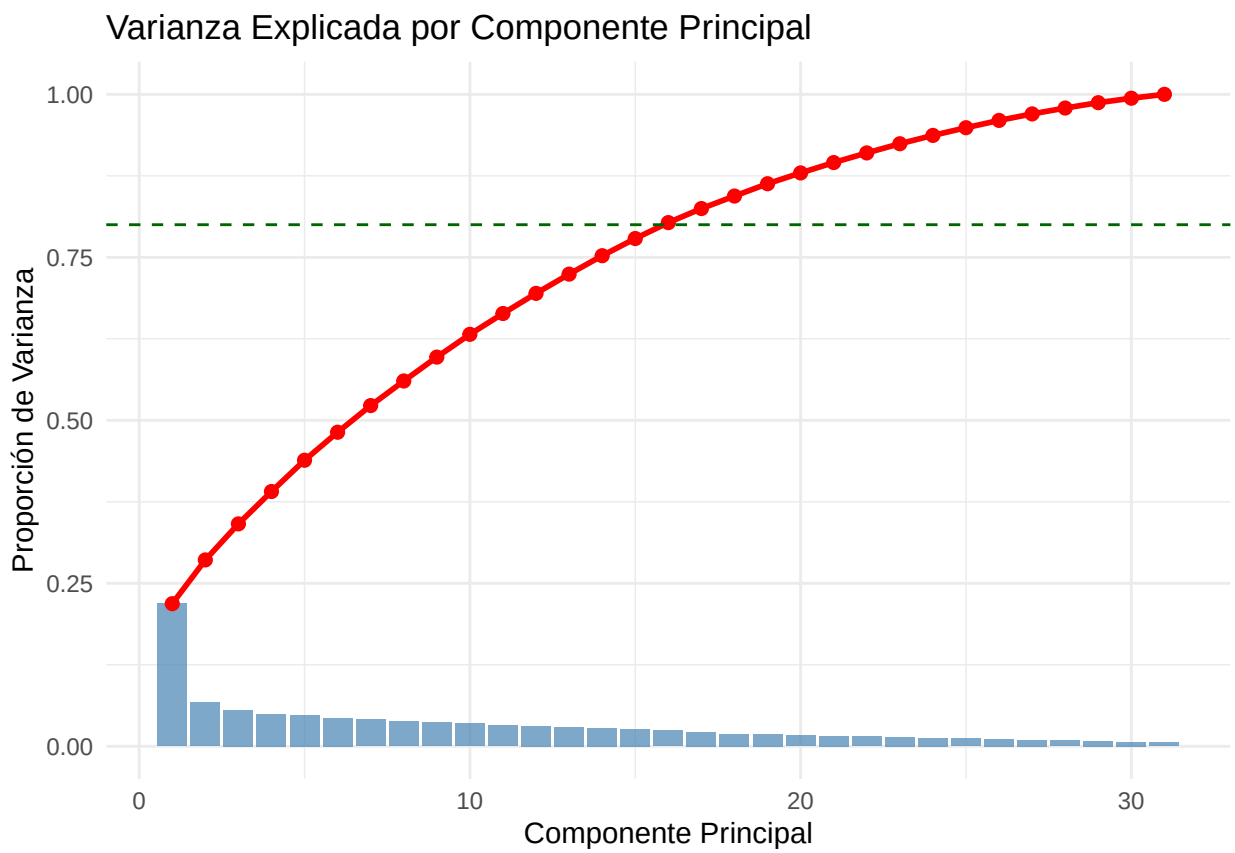
PCA SOBRE RENDIMIENTOS

```r
pca_result <- prcomp(mat_rendimientos, scale. = TRUE, center = TRUE)

# Varianza explicada
var_explicada    <- pca_result$sdev^2 / sum(pca_result$sdev^2)
var_acumulada    <- cumsum(var_explicada)

# Gráfico de varianza explicada (scree plot)
df_var <- data.frame(
  PC          = 1:length(var_explicada),
  Individual  = var_explicada,
  Acumulada   = var_acumulada
)
```

Gráfica de varianza explicada por componentes principales

```r
ggplot(df_var, aes(x = PC)) +
  geom_bar(aes(y = Individual), stat = "identity", fill = "steelblue", alpha = 0.7) +
  geom_line(aes(y = Acumulada), color = "red", size = 1) +
  geom_point(aes(y = Acumulada), color = "red", size = 2) +
  geom_hline(yintercept = 0.80, linetype = "dashed", color = "darkgreen") +
  labs(title = "Varianza Explicada por Componente Principal",
       x = "Componente Principal", y = "Proporción de Varianza") +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Componentes principales necesarias

```
# Número de PCs que explican >= 80% de la varianza
n_pcs <- which(var_acumulada >= 0.80)[1]
cat("PCs necesarias para explicar 80% de varianza:", n_pcs, "\n")
```

```
## PCs necesarias para explicar 80% de varianza: 16
```

```
# Cargas (loadings) de los primeros PCs
loadings_df <- as.data.frame(pca_result$rotation[, 1:min(n_pcs, 5)])
loadings_df$accion <- nombres_originales
print(loadings_df)
```

```
##                       PC1          PC2          PC3         PC4          PC5
## rendimiento_1  0.23249767  0.101536993 -0.160015996  0.25428986 -0.097388631
## rendimiento_2  0.21742706 -0.129898332  0.036868639 -0.03850844 -0.045525964
## rendimiento_3  0.23236079 -0.131175477  0.097287135 -0.05020308 -0.190596716
## rendimiento_4  0.19736660  0.437599777  0.154443735 -0.02224779 -0.036085913
## rendimiento_5  0.21220889 -0.108586079  0.283707532  0.09508748  0.354689599
## rendimiento_6  0.21720454 -0.106017980  0.072484229  0.35532180 -0.026958334
## rendimiento_7  0.09876466  0.326936306 -0.412951934 -0.02510253  0.238567280
## rendimiento_8  0.18780129  0.032330024 -0.001901165 -0.20358611  0.098010289
## rendimiento_9  0.20708584 -0.171553947 -0.012092312 -0.19742643 -0.283494248
## rendimiento_10 0.08726117 -0.046697653 -0.336462356 -0.11884796 -0.047823469
## rendimiento_11 0.15554264 -0.003494242  0.150311559  0.17237652 -0.191985821
```

```
## rendimiento_12 0.17573174  0.453181764  0.089860848 -0.18893864 -0.131519082
## rendimiento_13 0.22899332 -0.015469553 -0.006295858  0.29693158 -0.193331708
## rendimiento_14 0.16201983 -0.085703915 -0.075662400  0.01473803  0.004841757
## rendimiento_15 0.07644959 -0.026386137 -0.411525249 -0.09007753  0.155672447
## rendimiento_16 0.21636210 -0.158152340 -0.058883879  0.07904873 -0.088275693
## rendimiento_17 0.22743874 -0.164301875  0.010314440 -0.09181716  0.144010945
## rendimiento_18 0.13153553 -0.085070146  0.204369253 -0.15262360  0.165294236
## rendimiento_19 0.23989453 -0.030851111 -0.170389533  0.25283871 -0.169106793
## rendimiento_20 0.18372307 -0.075158242 -0.164126107 -0.08255868  0.039773645
## rendimiento_21 0.14774173 -0.035208716  0.122013604  0.05027875 -0.047836783
## rendimiento_22 0.07550001  0.166135108 -0.203306788  0.12503667  0.224136065
## rendimiento_23 0.13358299 -0.152172980 -0.371579400 -0.13263638 -0.191042438
## rendimiento_24 0.19111150  0.374188729  0.142744611 -0.22675335 -0.155428929
## rendimiento_25 0.10091319 -0.072971505 -0.065491665 -0.19593826  0.342473655
## rendimiento_26 0.18911752  0.079323794  0.138187772  0.19136919  0.297362979
## rendimiento_27 0.12057485  0.237546686 -0.007336363 -0.09989978 -0.111724154
## rendimiento_28 0.22635621  0.016049222  0.010320560  0.16875428  0.364933666
## rendimiento_29 0.08855262 -0.174055238  0.118825639 -0.34335437  0.076390256
## rendimiento_30 0.17875304 -0.055242863  0.101434469 -0.26415390  0.029582829
## rendimiento_31 0.19448224 -0.168420635  0.019219724 -0.20352713 -0.044935523
##                       accion
## rendimiento_1          AC.MX
## rendimiento_2       ALSEA.MX
## rendimiento_3        AMXB.MX
## rendimiento_4       ASURB.MX
## rendimiento_5     BBAJIOO.MX
## rendimiento_6      BIMBOA.MX
## rendimiento_7      BOLSAA.MX
## rendimiento_8    CEMEXCPO.MX
## rendimiento_9    CHDRAUIB.MX
## rendimiento_10     CUERVO.MX
## rendimiento_11   FEMSAUBD.MX
## rendimiento_12       GAPB.MX
## rendimiento_13   GCARSOA1.MX
## rendimiento_14        GCC.MX
## rendimiento_15    GENTERA.MX
## rendimiento_16   GFINBURO.MX
## rendimiento_17   GFNORTEO.MX
## rendimiento_18   GMEXICOB.MX
## rendimiento_19     GRUMAB.MX
## rendimiento_20    KIMBERA.MX
## rendimiento_21       LABB.MX
## rendimiento_22 LIVEPOLC-1.MX
## rendimiento_23    MEGACPO.MX
## rendimiento_24       OMAB.MX
## rendimiento_25      ORBIA.MX
## rendimiento_26     PINFRA.MX
## rendimiento_27          Q.MX
## rendimiento_28         RA.MX
## rendimiento_29 TLEVISACPO.MX
## rendimiento_30      VESTA.MX
## rendimiento_31     WALMEX.MX
```

Biplot de las primeras 2 componentes

```r
scores_df <- as.data.frame(pca_result$x[, 1:2])

fviz_pca_var(pca_result,
             col.var    = "contrib",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel      = TRUE,
             title      = "PCA - Variables (Acciones)") +
  theme_minimal()
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <e2>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
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## conversion failure on 'PCA - Variables (Acciones)' in 'mbcsToSbcs': dot
## substituted for <93>
```

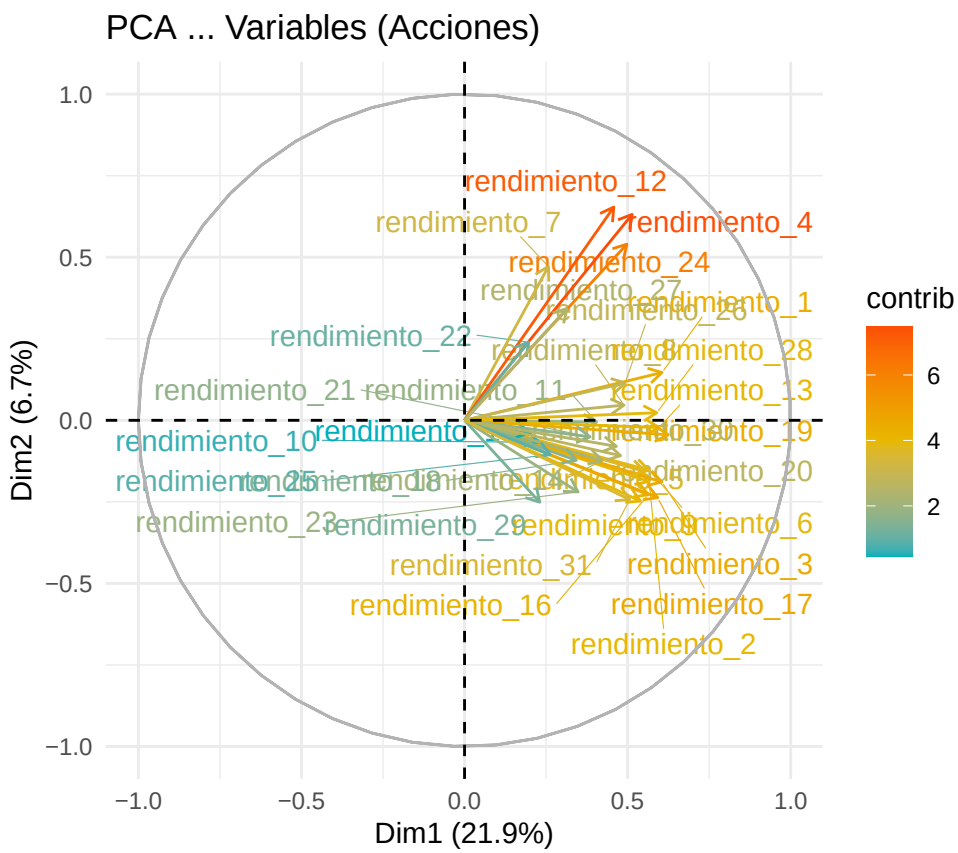

CLUSTERING SOBRE SCORES PCA

```
# Usamos los scores de las n_pcs componentes para clusterizar
scores_pca <- pca_result$x[, 1:n_pcs]

# Clustering sobre las ACCIONES (transponer: acciones en filas)
loadings_pca <- pca_result$rotation[, 1:n_pcs]   # filas = acciones

# Determinar número óptimo de clusters (método del codo)
set.seed(123)
fviz_nbclust(loadings_pca, kmeans, method = "wss",
             k.max = 30) +
  labs(title = "Método del Codo - Número Óptimo de Clusters") +
  theme_minimal()
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Método del Codo - Número Óptimo de Clusters' in
## 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
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## 'mbcsToSbcs': dot substituted for <80>

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## 'mbcsToSbcs': dot substituted for <93>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Método del Codo - Número Óptimo de Clusters' in
## 'mbcsToSbcs': dot substituted for <e2>

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## 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Método del Codo - Número Óptimo de Clusters' in
## 'mbcsToSbcs': dot substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Método del Codo - Número Óptimo de Clusters' in
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```

```
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## conversion failure on 'Método del Codo - Número Óptimo de Clusters' in
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```
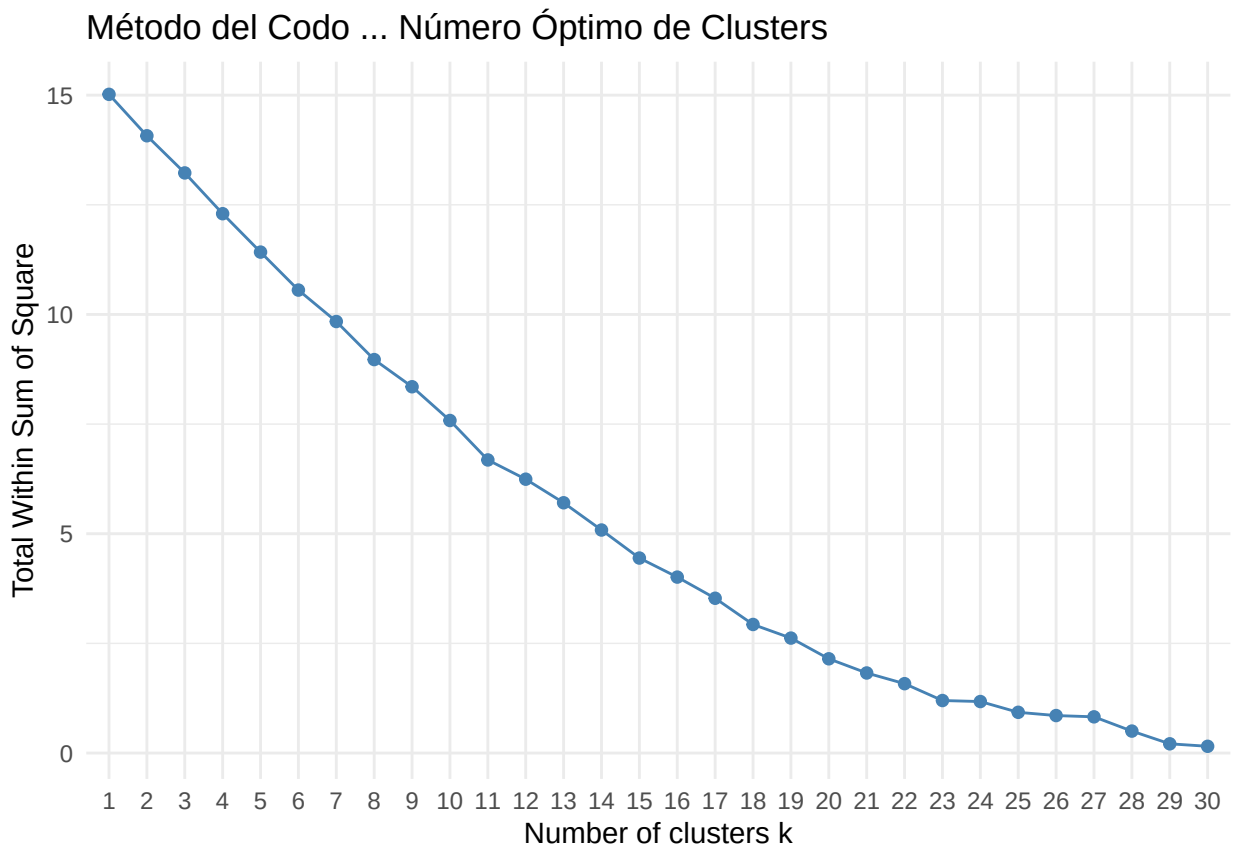




```r
# Número de clusters (ajusta según el codo observado)
k_optimo <- 4

set.seed(123)
km_result <- kmeans(loadings_pca, centers = k_optimo, nstart = 25, iter.max = 100)

# Asignar cluster a cada acción
df_clusters <- data.frame(
  accion_generica = nombres_genericos,
  accion          = nombres_originales,
  cluster         = km_result$cluster
)

cat("\nAcciones por cluster:\n")
```

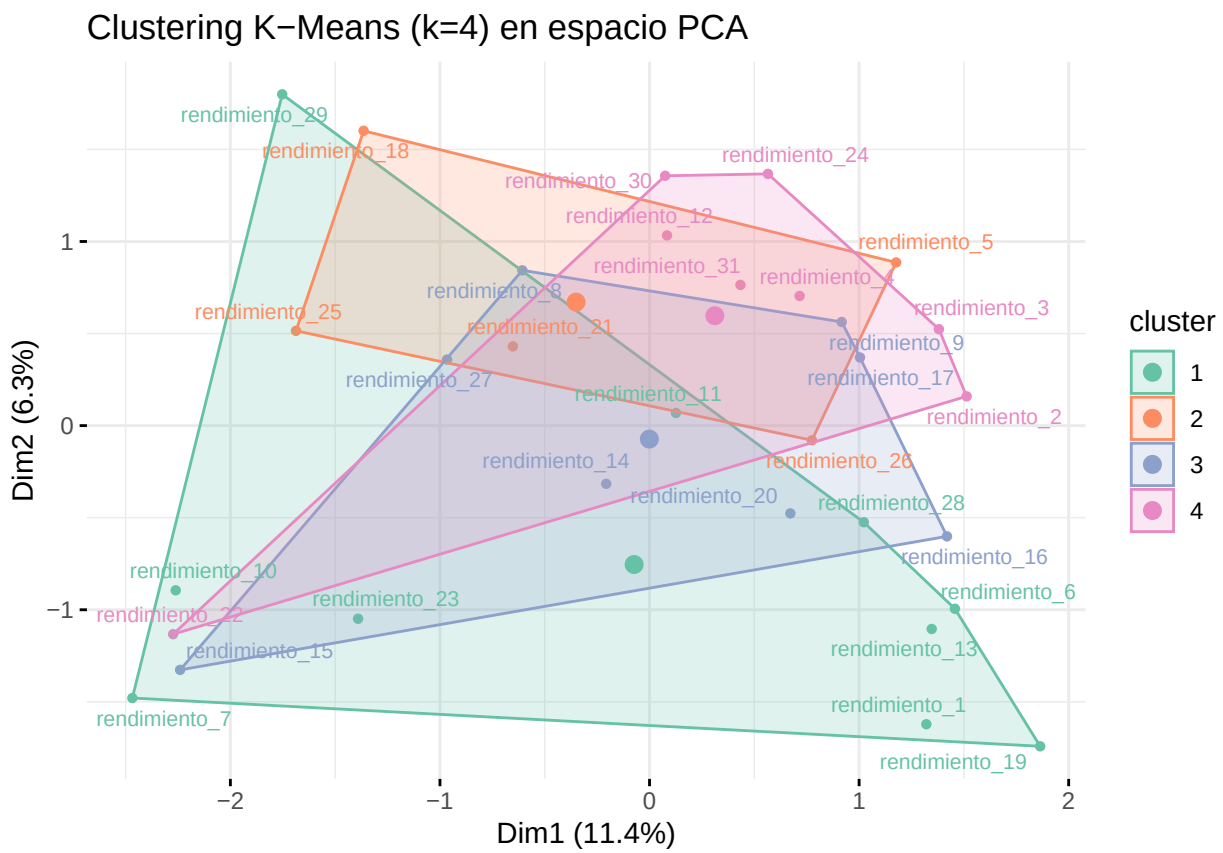
```
##
## Acciones por cluster:
```

```r
print(df_clusters %>% arrange(cluster))
```

```
##                accion_generica        accion cluster
## rendimiento_1         accion_1         AC.MX       1
## rendimiento_6         accion_6     BIMBOA.MX       1
## rendimiento_7         accion_7     BOLSAA.MX       1
```

```
## rendimiento_10       accion_10     CUERVO.MX       1
## rendimiento_11       accion_11   FEMSAUBD.MX       1
## rendimiento_13       accion_13   GCARSOA1.MX       1
## rendimiento_19       accion_19     GRUMAB.MX       1
## rendimiento_23       accion_23    MEGACPO.MX       1
## rendimiento_28       accion_28         RA.MX       1
## rendimiento_29       accion_29 TLEVISACPO.MX       1
## rendimiento_5         accion_5    BBAJIOO.MX       2
## rendimiento_18       accion_18   GMEXICOB.MX       2
## rendimiento_21       accion_21       LABB.MX       2
## rendimiento_25       accion_25      ORBIA.MX       2
## rendimiento_26       accion_26     PINFRA.MX       2
## rendimiento_8         accion_8   CEMEXCPO.MX       3
## rendimiento_9         accion_9   CHDRAUIB.MX       3
## rendimiento_14       accion_14        GCC.MX       3
## rendimiento_15       accion_15    GENTERA.MX       3
## rendimiento_16       accion_16   GFINBURO.MX       3
## rendimiento_17       accion_17   GFNORTEO.MX       3
## rendimiento_20       accion_20    KIMBERA.MX       3
## rendimiento_27       accion_27          Q.MX       3
## rendimiento_2         accion_2      ALSEA.MX       4
## rendimiento_3         accion_3       AMXB.MX       4
## rendimiento_4         accion_4      ASURB.MX       4
## rendimiento_12       accion_12       GAPB.MX       4
## rendimiento_22       accion_22 LIVEPOLC-1.MX       4
## rendimiento_24       accion_24       OMAB.MX       4
## rendimiento_30       accion_30      VESTA.MX       4
## rendimiento_31       accion_31     WALMEX.MX       4
```

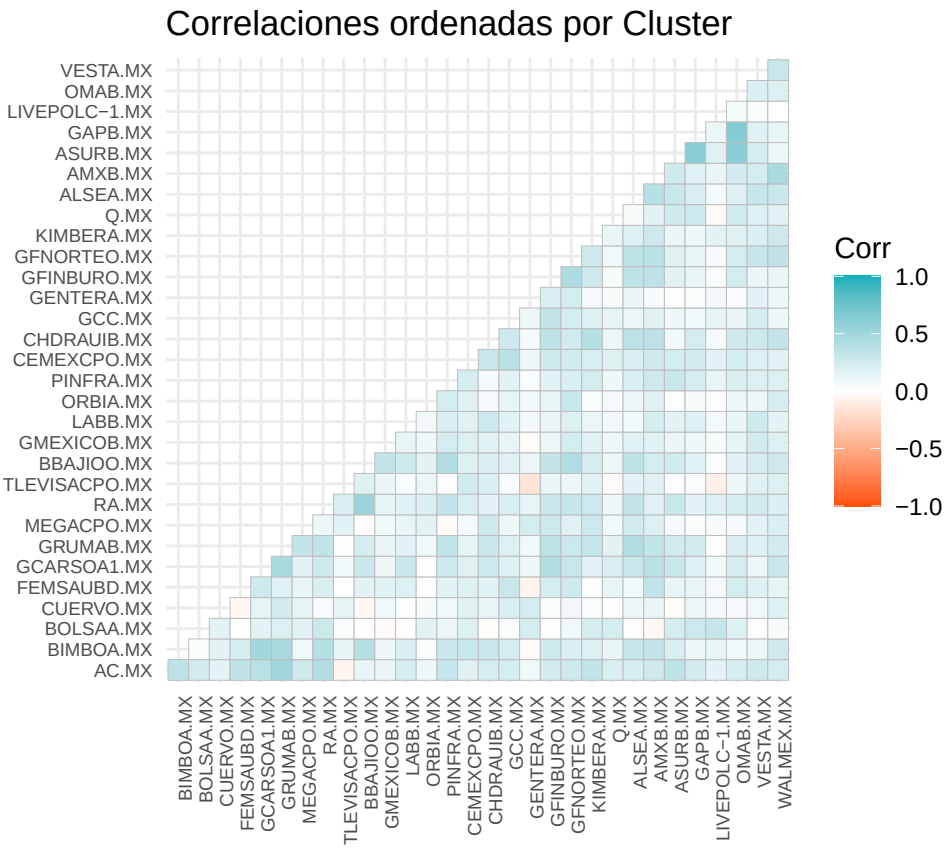
```r
# Visualización de clusters en espacio PCA
fviz_cluster(km_result,
             data          = loadings_pca,
             labelsize     = 8,
             repel         = TRUE,
             ellipse.type  = "convex",
             palette       = "Set2",
             ggtheme       = theme_minimal()) +
  labs(title = paste0("Clustering K-Means (k=", k_optimo, ") en espacio PCA")) +
  scale_shape_manual(values = rep(16, k_optimo))
```




```r
# Mapa de calor de correlaciones coloreado por cluster
orden_cluster <- df_clusters %>% arrange(cluster) %>% pull(accion_generica)
idx_orden     <- match(orden_cluster, nombres_genericos)
cor_mat       <- cor(mat_rendimientos)[idx_orden, idx_orden]
colnames(cor_mat) <- rownames(cor_mat) <- df_clusters$accion[match(orden_cluster, df_clusters$accion_ge
```

Gráfico de correlaciones

```r
ggcorrplot(cor_mat,
           hc.order   = FALSE,
           type       = "lower",
           lab        = FALSE,
           tl.cex     = 7,
           colors     = c("#FC4E07", "white", "#00AFBB"),
           title      = "Correlaciones ordenadas por Cluster") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



SELECCIÓN DE REPRESENTANTE POR CLUSTER

```
representantes <- df_clusters %>%
  left_join(df_stats %>% mutate(accion_generica = nombres_genericos),
            by = "accion_generica") %>%
  group_by(cluster) %>%
  slice_max(sharpe, n = 1, with_ties = FALSE) %>%
  ungroup()

# DEBUG: ver qué columnas existen
cat("Columnas en representantes:\n")
```

```
## Columnas en representantes:
```

```
print(names(representantes))
```

```
## [1] "accion_generica" "accion.x"        "cluster"         "accion.y"
## [5] "media_diaria"    "desv_diaria"     "sharpe"
```

```
cat("\nRepresentante seleccionado por cluster:\n")
```

```
##
## Representante seleccionado por cluster:
```

```r
print(representantes %>% dplyr::select(cluster, accion.x, sharpe))
```

```
## # A tibble: 4 x 3
##   cluster accion.x      sharpe
##     <int> <chr>          <dbl>
## 1       1 TLEVISACPO.MX  0.846
## 2       2 GMEXICOB.MX    2.20
## 3       3 CEMEXCPO.MX    3.18
## 4       4 AMXB.MX        1.30
```

```r
acciones_seleccionadas <- representantes$accion_generica
nombres_seleccionados  <- representantes$accion.x
n_sel <- length(acciones_seleccionadas)
```

PORTAFOLIO ÓPTIMO (MÁXIMO SHARPE) CON QUADPROG

```r
# Sub-matrices para acciones seleccionadas
idx_sel     <- match(acciones_seleccionadas, nombres_genericos)
medias_sel  <- medias[idx_sel]
cov_sel     <- covarianza[idx_sel, idx_sel]

max_sharpe_portfolio <- function(mu, Sigma, rf, n_sim = 5000) {
  n <- length(mu)

  set.seed(42)
  pesos_sim  <- matrix(runif(n_sim * n), ncol = n)
  pesos_sim  <- pesos_sim / rowSums(pesos_sim)

  ret_sim    <- pesos_sim %*% mu * 252

  # Calcular varianza fila por fila para no agotar memoria
  riesgo_sim <- sqrt(sapply(1:n_sim, function(i) {
    p <- pesos_sim[i, ]
    t(p) %*% Sigma %*% p
  })) * sqrt(252)

  sharpe_sim <- (ret_sim - rf * 252) / riesgo_sim

  mejor_idx  <- which.max(sharpe_sim)
  pesos_mc   <- pesos_sim[mejor_idx, ]
  ret_obj    <- as.numeric(ret_sim[mejor_idx]) / 252

  # Refinamiento con quadprog
  Dmat <- 2 * Sigma
  dvec <- rep(0, n)
  Amat <- cbind(rep(1, n), mu, diag(n))
  bvec <- c(1, ret_obj, rep(0, n))
  meq  <- 1

  sol <- tryCatch(
    solve.QP(Dmat, dvec, Amat, bvec, meq = meq),
    error = function(e) NULL
```

```r
  )

  if (!is.null(sol)) {
    pesos_opt <- sol$solution
    pesos_opt[pesos_opt < 1e-6] <- 0
    pesos_opt <- pesos_opt / sum(pesos_opt)
  } else {
    pesos_opt <- pesos_mc
  }

  ret_opt    <- sum(pesos_opt * mu) * 252
  riesgo_opt <- sqrt(t(pesos_opt) %*% Sigma %*% pesos_opt) * sqrt(252)
  sharpe_opt <- (ret_opt - rf * 252) / riesgo_opt

  list(
    pesos    = pesos_opt,
    retorno  = ret_opt,
    riesgo   = as.numeric(riesgo_opt),
    sharpe   = as.numeric(sharpe_opt),
    frontera = data.frame(retorno = as.numeric(ret_sim),
                          riesgo  = riesgo_sim,
                          sharpe  = as.numeric(sharpe_sim))
  )
}

resultado <- max_sharpe_portfolio(medias_sel, cov_sel, r_f)
```

```r
# Tabla de pesos
df_pesos <- data.frame(
  Accion  = nombres_seleccionados,
  Cluster = representantes$cluster,
  Peso    = round(resultado$pesos * 100, 2)
) %>% filter(Peso > 0.01)

cat("\n=== PORTAFOLIO ÓPTIMO (Máximo Sharpe) ===\n")
```

```
##
## === PORTAFOLIO ÓPTIMO (Máximo Sharpe) ===
```

```r
print(df_pesos)
```

```
##        Accion Cluster  Peso
## 1 GMEXICOB.MX       2 27.65
## 2 CEMEXCPO.MX       3 62.51
## 3     AMXB.MX       4  9.84
```

```r
cat(sprintf("\nRendimiento anualizado : %.2f%%\n", resultado$retorno * 100))
```

```
##
## Rendimiento anualizado : 83.10%
```

```r
cat(sprintf("Riesgo anualizado  : %.2f%%\n", resultado$riesgo  * 100))
```

```
## Riesgo anualizado  : 23.16%
```

```r
cat(sprintf("Sharpe ratio       : %.4f\n",   resultado$sharpe))
```

```
## Sharpe ratio       : 3.5861
```

VISUALIZACIONES FINALES

```r
#Frontera eficiente + portafolio óptimo
frontera <- resultado$frontera

ggplot(frontera, aes(x = riesgo * 100, y = retorno * 100, color = sharpe)) +
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_gradient(low = "blue", high = "red", name = "Sharpe") +
  geom_point(aes(x = resultado$riesgo * 100, y = resultado$retorno * 100),
             color = "gold", size = 5, shape = 18) +
  annotate("text", x = resultado$riesgo * 100, y = resultado$retorno * 100,
           label = "Máx. Sharpe", vjust = -1, fontface = "bold", color = "black") +
  labs(title = "Frontera Eficiente - Portafolio PCA + Clustering",
       x     = "Rendimiento (%)",
       y     = "Riesgo (%)") +
  theme_minimal()
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Frontera Eficiente - Portafolio PCA + Clustering' in
## 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Frontera Eficiente - Portafolio PCA + Clustering' in
## 'mbcsToSbcs': dot substituted for <80>

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## conversion failure on 'Frontera Eficiente - Portafolio PCA + Clustering' in
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## 'mbcsToSbcs': dot substituted for <e2>
```

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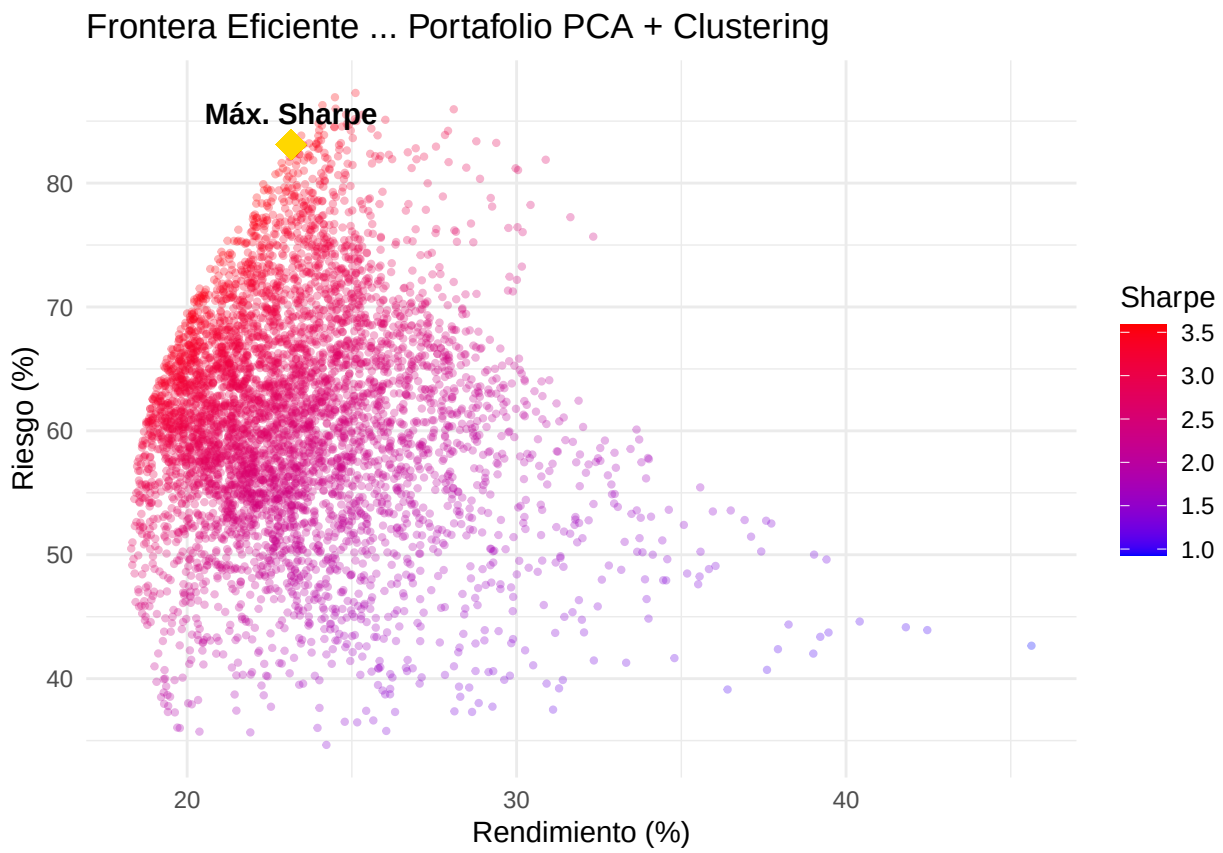
## 'mbcsToSbcs': dot substituted for <93>
```



Métricas de desempeño

Rendimiento diario del portafolio óptimo

```r
# Rendimientos diarios solo de las acciones seleccionadas
rendimientos_sel <- mat_rendimientos[, idx_sel]

# Vector de pesos óptimos
pesos_opt <- resultado$pesos

# Rendimiento diario del portafolio
rend_portafolio <- rendimientos_sel %*% pesos_opt

# Convertir a vector
rend_portafolio <- as.numeric(rend_portafolio)

# Crear dataframe con fechas
df_portafolio <- data.frame(
  Fecha = df_rendims_acciones$Fecha,
  Rendimiento = rend_portafolio
)

head(df_portafolio)
```

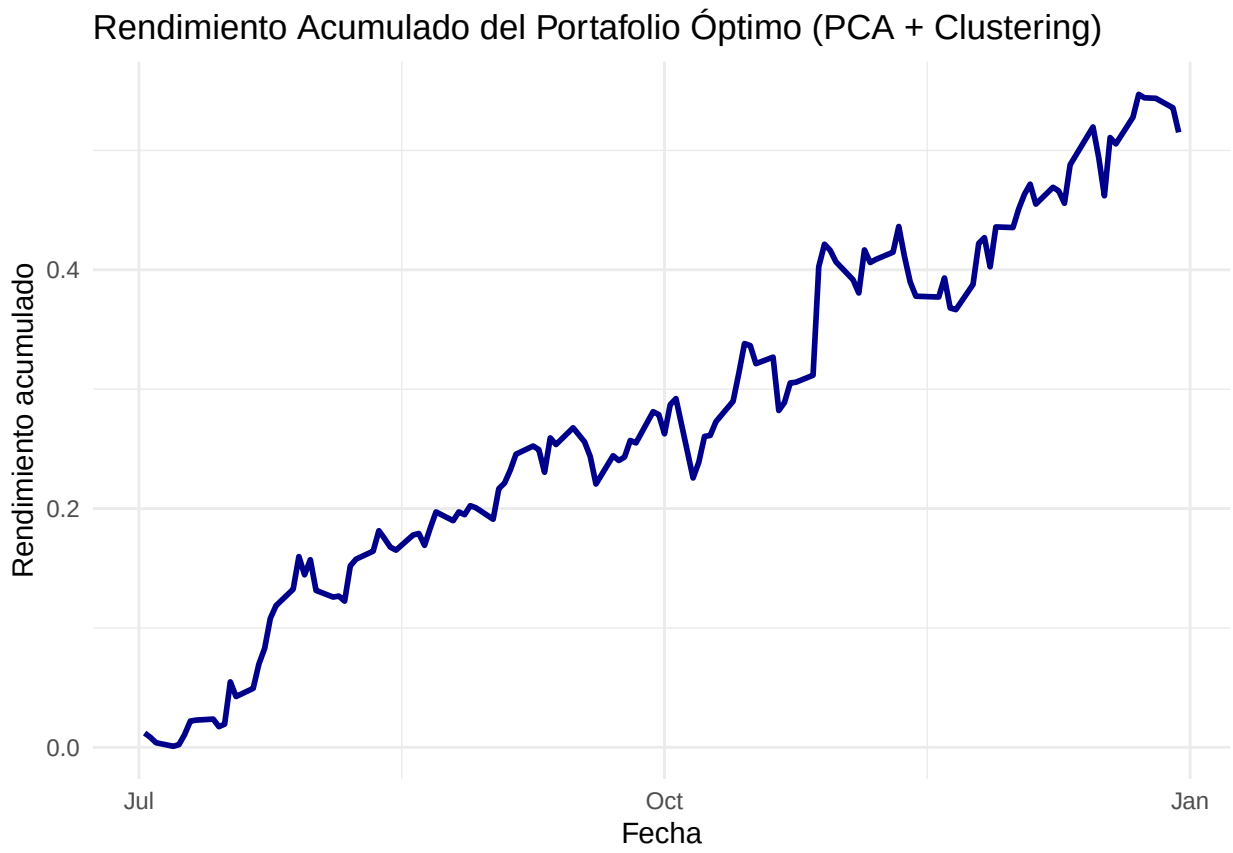
```
##        Fecha  Rendimiento
## 1 2025-07-02  0.011887754
## 2 2025-07-03 -0.003540866
## 3 2025-07-04 -0.004513565
## 4 2025-07-07 -0.002989138
## 5 2025-07-08  0.001350759
## 6 2025-07-09  0.008442739
```

```r
# Rendimiento acumulado (usando log-returns)
df_portafolio$Rend_Acumulado <- exp(cumsum(df_portafolio$Rendimiento)) - 1

# Último rendimiento acumulado total
rendimiento_total <- tail(df_portafolio$Rend_Acumulado, 1)

cat("Rendimiento acumulado total del portafolio:",
    round(rendimiento_total * 100, 4), "%\n")
```

```
## Rendimiento acumulado total del portafolio: 51.5105 %
```

Gráfica del rendimiento acumulado

```r
ggplot(df_portafolio, aes(x = Fecha, y = Rend_Acumulado)) +
  geom_line(size = 1, color = "darkblue") +
  labs(title = "Rendimiento Acumulado del Portafolio Óptimo (PCA + Clustering)",
       x = "Fecha",
       y = "Rendimiento acumulado") +
  theme_minimal()
```



Ratios del portafolio Π

Sharp-Ratio

```
# Media y desviación diaria del portafolio
media_port   <- mean(rend_portafolio)
desv_port    <- sd(rend_portafolio)

# Sharpe anual (360 días)
sharpe_port <- (media_port * 360 - r_f * 360) / (desv_port * sqrt(360))

cat("Sharpe Ratio anual del portafolio óptimo:",
    round(sharpe_port, 6), "\n")
```

```
## Sharpe Ratio anual del portafolio óptimo: 4.286161
```

Treynor-ratio

```
# Descargar IPC
ipc_data <- BatchGetSymbols(tickers = "^MXX",
                            first.date = primera_fecha,
                            last.date = ultima_fecha,
                            freq.data = frecuencia)
```

```
##
## Running BatchGetSymbols for:
##    tickers =^MXX
##    Downloading data for benchmark ticker
## ^GSPC | yahoo (1|1) | Found cache file
## ^MXX | yahoo (1|1) | Not Cached | Saving cache - Got 98% of valid prices | Nice!
```

```r
ipc_precios <- ipc_data$df.tickers

df_ipc <- ipc_precios %>%
  dplyr::select(ref.date, price.adjusted) %>%
  arrange(ref.date)

# Log-rendimientos IPC
ipc_rend <- diff(log(df_ipc$price.adjusted))

# Alinear fechas
ipc_rend <- ipc_rend[!is.na(ipc_rend)]

df_mercado <- data.frame(
  Fecha = df_rendims_acciones$Fecha,
  Rend_IPC = ipc_rend
)
```

Se calcula la β

```r
# Alinear longitudes por seguridad
min_len <- min(length(rend_portafolio), length(df_mercado$Rend_IPC))

rp <- rend_portafolio[1:min_len]
rm <- df_mercado$Rend_IPC[1:min_len]

# Beta = Cov(Rp, Rm) / Var(Rm)
beta_port <- cov(rp, rm) / var(rm)

cat("Beta del portafolio:", round(beta_port, 6), "\n")
```

```
## Beta del portafolio: 1.205851
```

```r
# Media diaria del portafolio
media_port <- mean(rp)

# Treynor anualizado
treynor_port <- ((media_port * 360) - (r_f * 360)) / beta_port

cat("Treynor Ratio anualizado del portafolio:",
    round(treynor_port, 6), "\n")
```

```
## Treynor Ratio anualizado del portafolio: 0.983802
```

Sortino Ratio

```r
# Alinear datos (por seguridad)
min_len <- length(rend_portafolio)
rp <- rend_portafolio[1:min_len]

# Excess return diario
excess_ret <- rp - r_f

# Downside deviation (solo retornos negativos respecto a rf)
downside <- ifelse(excess_ret < 0, excess_ret, 0)

downside_dev <- sqrt(mean(downside^2))

# Sortino anualizado
sortino_port <- ((mean(rp) * 252) - (r_f * 252)) / (downside_dev * sqrt(252))

cat("Sortino Ratio anualizado del portafolio:",
    round(sortino_port, 6), "\n")
```

```
## Sortino Ratio anualizado del portafolio: 6.199068
```

##Gráfica comparativa

```r
# Rendimiento acumulado IPC (base 100)
df_mercado$Indice_IPC <- exp(cumsum(df_mercado$Rend_IPC))

# Asegurarnos que el portafolio también esté en base 100
df_portafolio$Indice_Portafolio <- exp(cumsum(df_portafolio$Rendimiento))
```

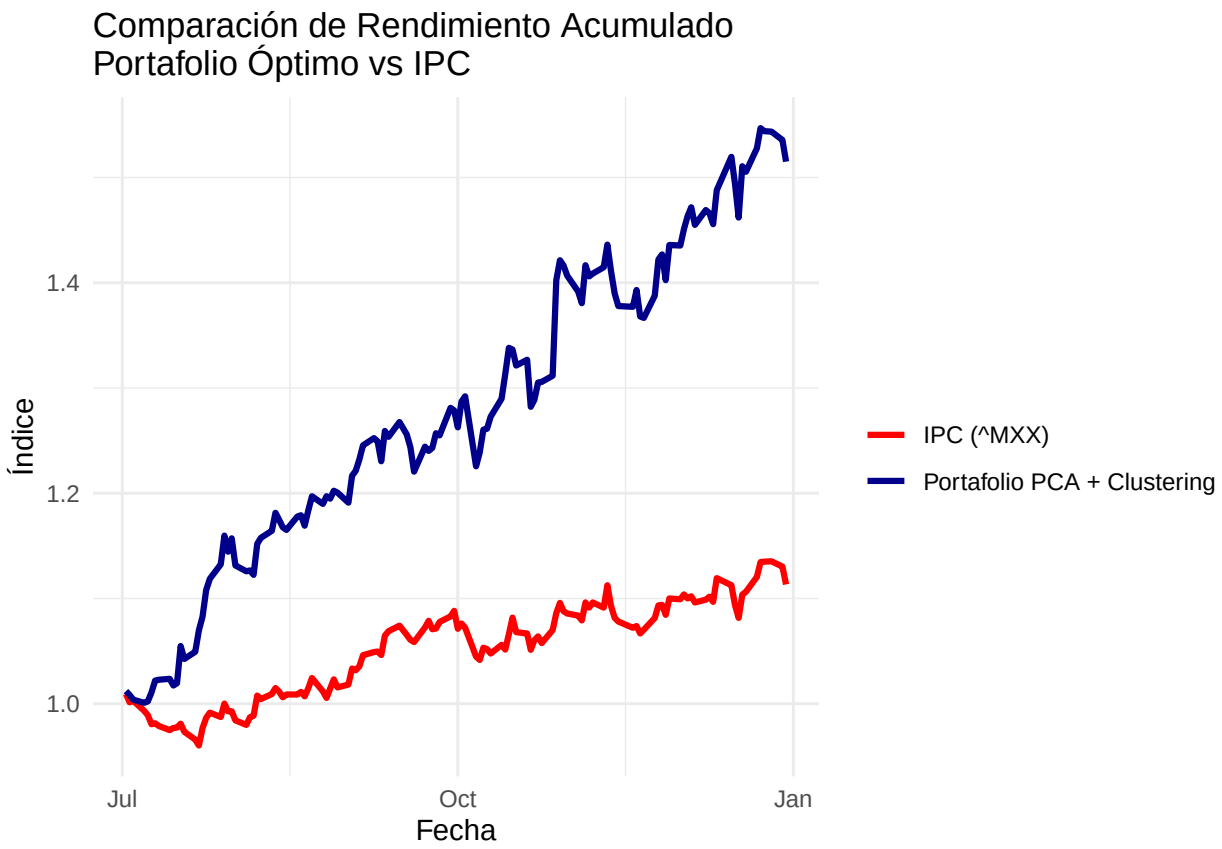
Unir el portafolio

```r
# Unir por fecha para garantizar alineación perfecta
df_comparacion <- df_portafolio %>%
  dplyr::select(Fecha, Indice_Portafolio) %>%
  inner_join(df_mercado %>%
               dplyr::select(Fecha, Indice_IPC),
             by = "Fecha")
```

Formato largo

```r
df_long <- df_comparacion %>%
  pivot_longer(cols = c(Indice_Portafolio, Indice_IPC),
               names_to = "Serie",
               values_to = "Valor")
```

Gráfica

```r
ggplot(df_long, aes(x = Fecha, y = Valor, color = Serie)) +
  geom_line(size = 1.1) +
  labs(title = "Comparación de Rendimiento Acumulado\nPortafolio Óptimo vs IPC",
       x = "Fecha",
       y = "Índice",
```

```
      color = "") +
  theme_minimal() +
  scale_color_manual(values = c("Indice_Portafolio" = "darkblue",
                                "Indice_IPC" = "red"),
                     labels = c("IPC (^MXX)", "Portafolio PCA + Clustering"
                                ))
```